SmartBed

MVP Specs slight introduction

Disclaimer

1. Minimum Viable Product (MVP) Status:

The specifications provided in this document are intended for the Minimum Viable Product (MVP) version of the Smart Bed. As an MVP, the components and features listed here are selected for flexible development purposes. It is important to note that these specifications are subject to change, improvement, and upgrade in subsequent versions as we gather user feedback and enhance the product's capabilities.

1. Variability of Component Quality:

The Smart Bed's components might exhibit a range of quality levels. While some parts may be sourced with cost-efficiency in mind and could be of cheaper quality, others may incorporate enterprise-level products known for their superior performance and durability. All components will undergo rigorous testing and adjustments to meet market price targets while ensuring satisfactory customer experiences.

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1. Concept ideas

- Introducing the Smart Bed, a cutting-edge sleep solution designed to transform your sleeping experience.
- Our goal is to create a bed that adapts to your needs, enhances comfort, and promotes overall well-being.
- With a range of innovative features, the Smart Bed is set to revolutionize the way you sleep.

1. Concept ideas





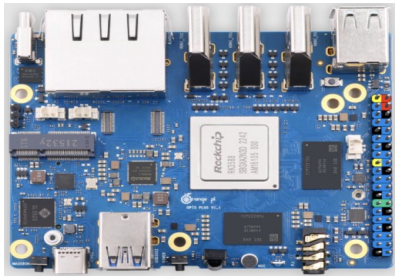
- It is a soundproof sleeping pod (chamber)
- Embodying elegance, luxury, and a futuristic aesthetic inspired by industry pioneers like Apple and Tesla.
- Provide a friendly, familiar, and safe feeling, ensuring our users feel at ease and well-rested.
- All sensors and collector/input devices are non-contact to the user and hiding from the interface

2. CPU

- At the heart of the Smart Bed is a powerful CPU that serves as its brain.
- The CPU efficiently processes data from various sensors, doing the AI stuffs and controls the bed's functionalities.
- It enables real-time adjustments to ensure an optimal sleeping environment.

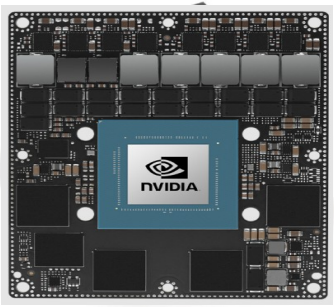
2. CPU











- Main CPU: Orange Pi 5 plus
  - Rockchip RK3588 8-core 64-bit processor
  - 16GB Ram (LPDDR4/4x)
  - Dual 8k output
  - 256GB eMMC storage
  - wifi/bluetooth/dual 2.5G Lan ports
  - 40 GPIOs
- Google Coral USB Stick
  - accelerate ML inferencing
  - 4 trillion operations (tera-operations) per second (TOPS)
- Nvidia Jetson Orin Module
  - accelerate computer visions
  - 275 TOPS

3. Tech stacks

Flutter

node

GO

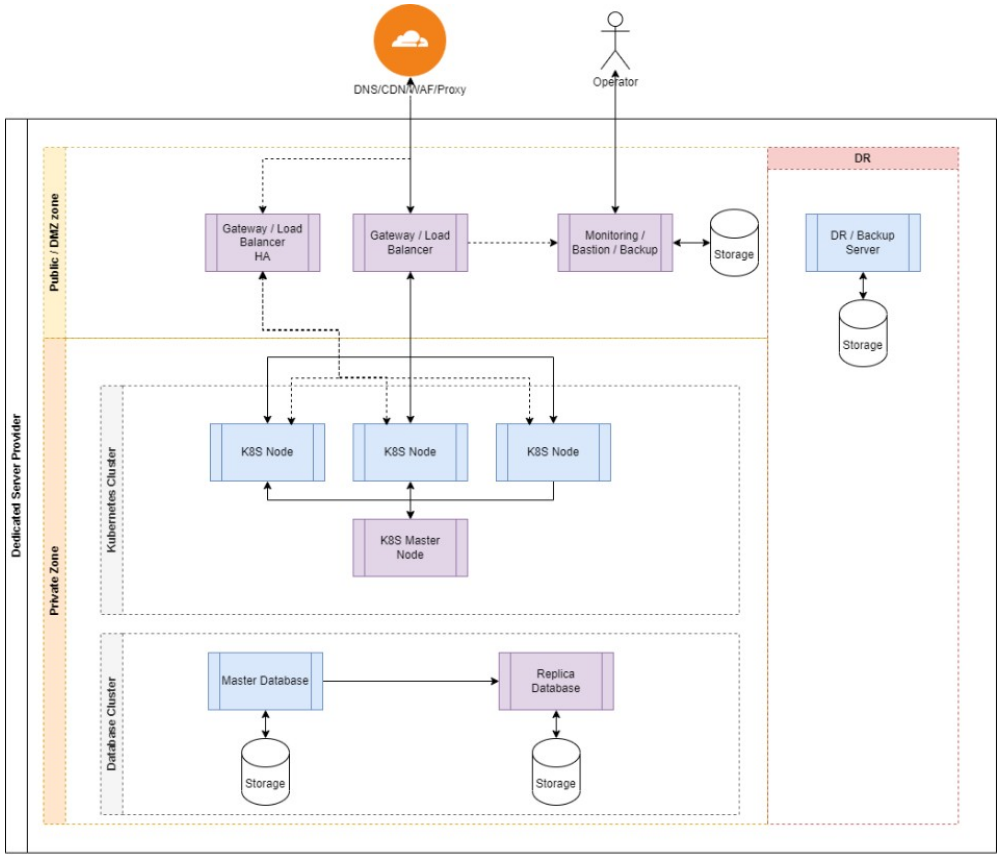
python

SOLIDITY

mongoDB

CockroachDB

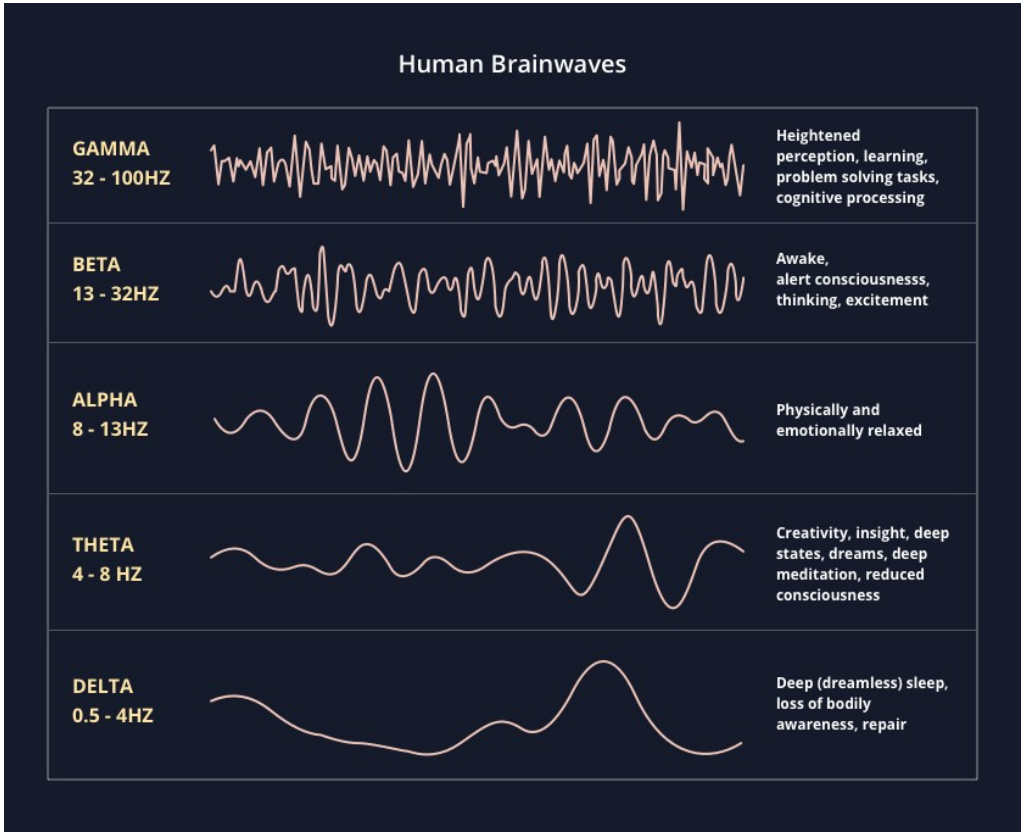
redis



4. Sensors, Camera, Micro

- Prioritizing user comfort and convenience, all sensors and input devices are discreetly hidden from the interface.
- All the input sources are modular, which means can be added and removed without harm to the bed operation.
- There are multiple of input source for a factor to ensure the stability and accuracy data/AI-algorithm.
- The SmartBed operates effortlessly, ensuring a seamless, non-intrusive experience.

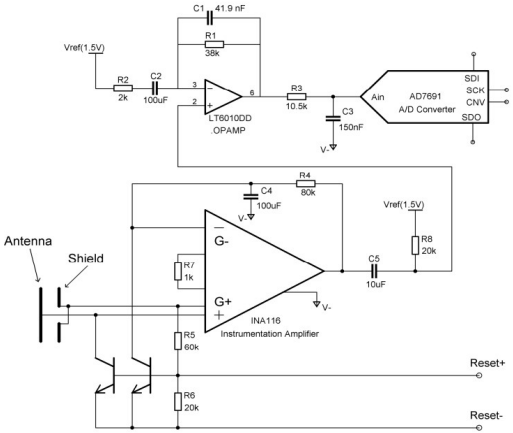
4.1. Brain-wave sensor (EEG, ECG)



- The Smart Bed comes equipped with a state-of-the-art non-contact brain-wave sensor.
- This non-intrusive technology monitors your brain activity during sleep, providing valuable insights into sleep patterns and quality.

4.1. Brain-wave sensor (EEG, ECG)





- [A Low-Noise, Non-Contact EEG/ECG Sensor]
- The measured input-referred noise, over 1-100 Hz frequency range, is 2μVrms at 0.2mm sensor distance, and 17μVrms at 3.2mm distance. Experiments coupling the sensor to the human scalp through hair and to chest through clothing produce clear EEG and ECG recorded sign
- A co nside the p



4.1. Brain-wave sensor (EEG, ECG)

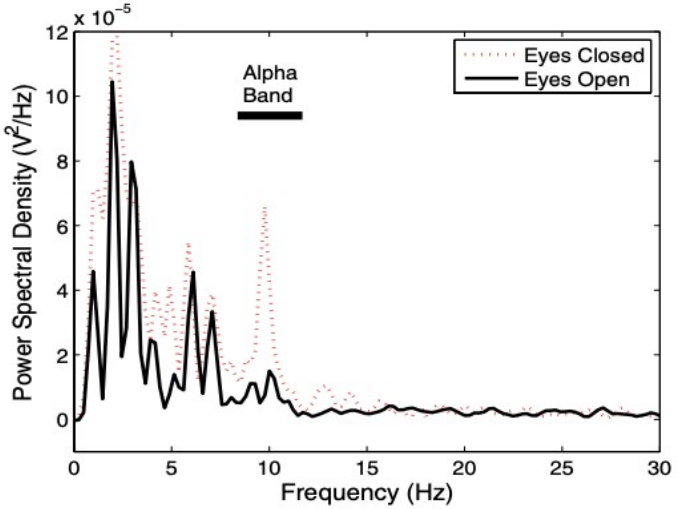


Fig. 5. Alpha waves. One sensor was place on the back of the subject's head and one was placed behind the ear as a reference. The subject closed eyes for 12 seconds and kept them open for 12 seconds. The power spectrum of the two blocks of data reveals that the power near 10 Hz, commonly called alpha waves, is bigger with the eyes closed. This is commonly observed in EEG research.

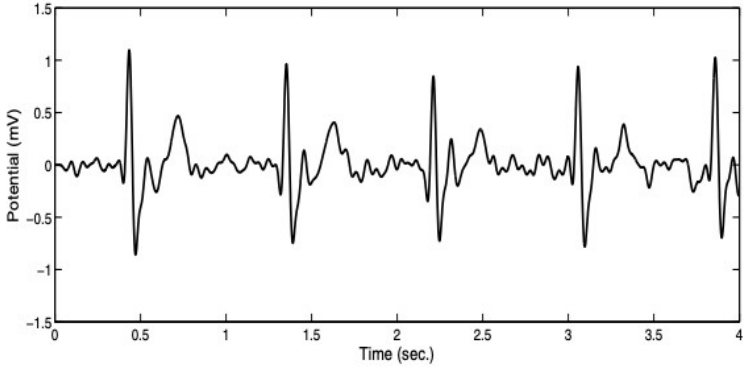
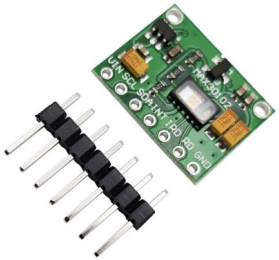


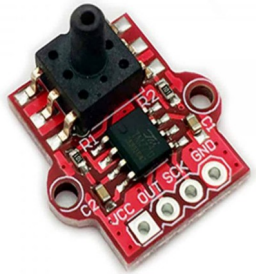
Fig. 6. ECG. A typical ECG voltage was measured through a T-shirt. Shown is the potential difference between two sensors near the heart.

4.2. Vital sign sensors

- Multiple living factor sensors detect vital signs and body movements throughout the night.
- These sensors offer personalized sleep analysis and contribute to intelligent adjustments for a better night's rest.
- The output of these sensors are:
    - Heart beat
    - SpO2
    - Blood pressure
    - Weight
    - Body temperature
    - Human presence
    - Breath pattern
    - Breath analyzer
    - Temperature and Humidity

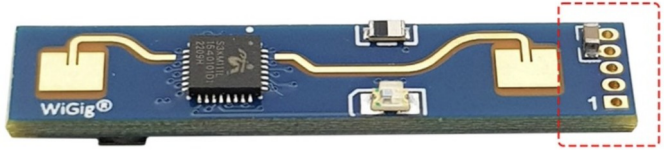
4.2. Vital sign sensors



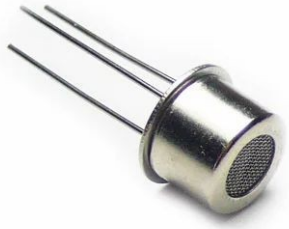








- Heartbeat & SpO2: MAX30102 - G4VK
- Blood pressure: MPS20N400D
- Weight sensor 150Kg: Loadcell 150
- Body thermal sensor: CJMCU-MLX90614 BCC
- Human presence & breath & heartbeat: HLK-LD2410B+BLE 24Ghz
- Breath analyzer: MQ303B Alcohol Gas sensor
- Temperature and Humidity Sensor: CJMCU-SHT10





4.3. Directional micro

- The directional microphone embedded in the bed can be controlled to captures and analyzes sounds from user body part (mouth, nose) and from ambient.
- The input source of sound from the microphone is placed an important weight in detection sleep apnea, snoring during sleep and also the requirement factor for AI-algorithms.
- The feedback from microphone helps adjusting soundscapes and white noise to create a serene sleeping environment, free from disruptions.

4.3. Directional micro

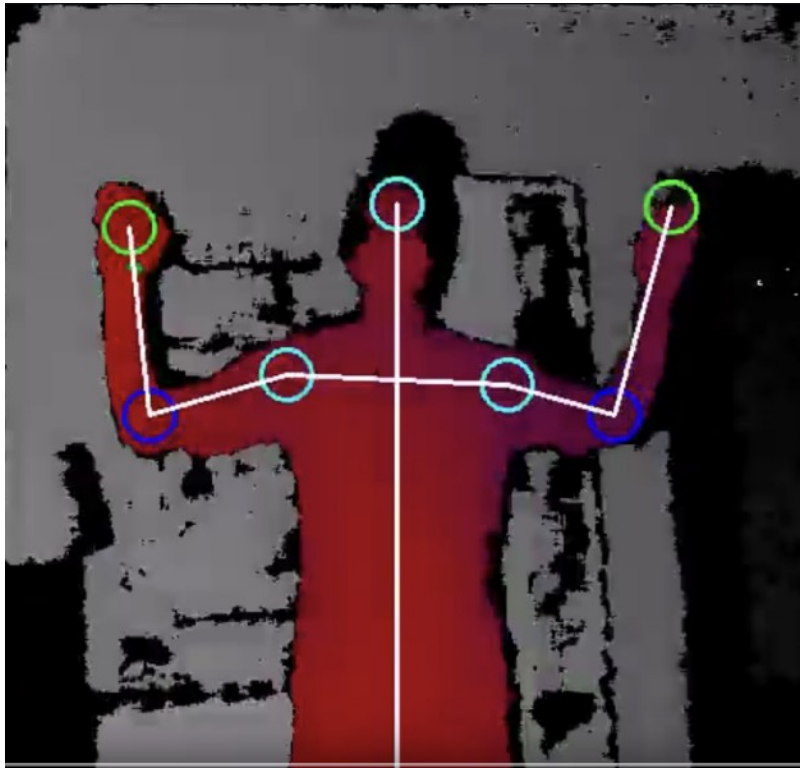


- LensGo LYM-DM300 - Directional Shotgun Microphone
  - 40-20KHZ
  - -38dB □ 3dB, 0dB = 1V / PA, 1KHZ
  - 78dB
  - 200 Ohms □ 30%

4.3. Camera & data privacy

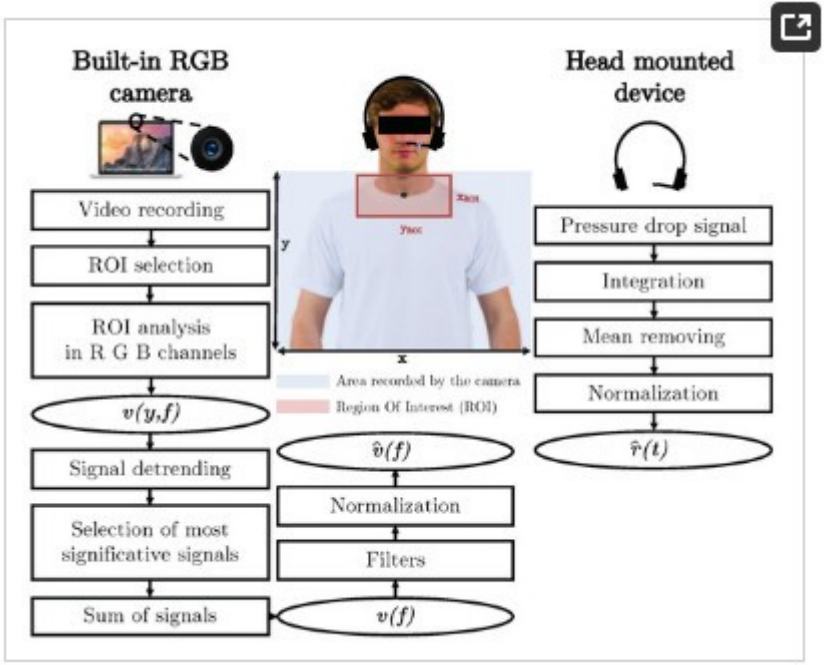
- The optional integrated camera can be used for sleep analysis and tracking sleep positions.
- We prioritize data privacy and offer features to disable or cover the camera as per user preferences.
- One of these could be used for camera input source: high speed accuracy camera, thermal camera, linear camera, RGB camera.
- All the camera stream will be hashed with local unique passkey and only tag-hashed data being save to the local storage for machine learning purpose. Those data can not be read by human or send out.

4.4.1 Sleep position detection



- Skeleton body detection and tracking using camera via computer vision
- From the real time data collected, we can use it to:
    - Detect body size, body height, skeleton marker, sleep position, body movement.
    - Adjusting the 10 kinematic points to fit any body size
    - Controlling 10 kinematic points to manipulating sleep position

4.4.2 Breathing Pattern and Respiratory Rate



- [Non-Contact Monitoring of Breathing Pattern and Respiratory Rate via RGB Signal Measurement]
- We will use directional microphones in the 4.3 to replace the head mounted device.

4.4.3 Blood Pressure

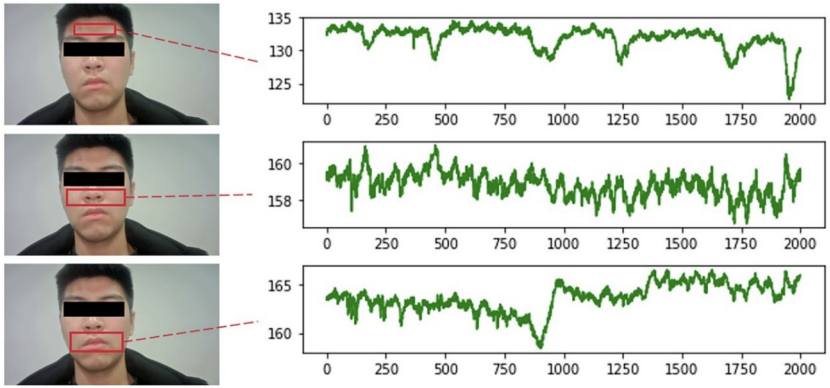
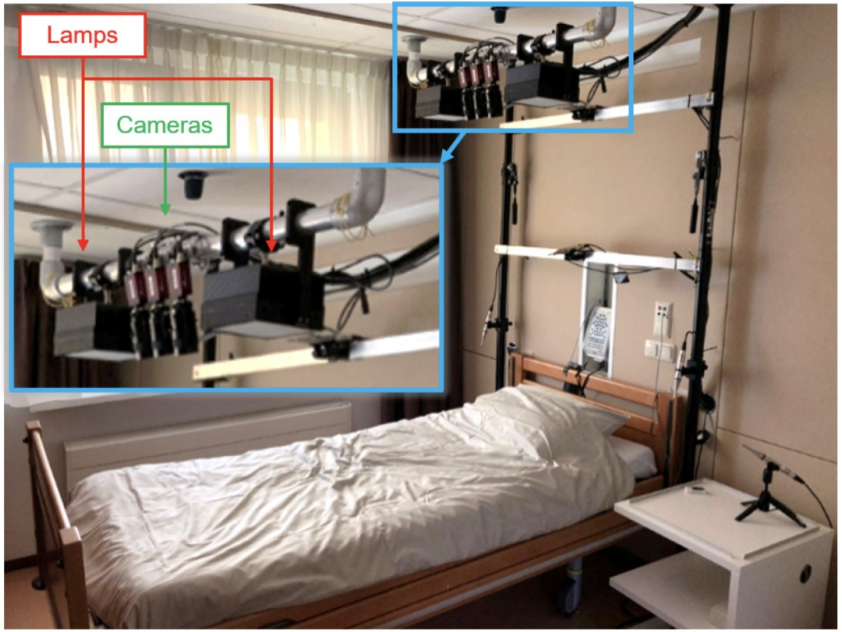


Figure 7. Comparison of different ROI signals.

- [Deep Learning-Based Non-Contact IPPG Signal Blood Pressure Measurement Research]
- A multi-stage deep learning blood pressure prediction model based on imaging photoplethysmography (IPPG) signals is proposed to achieve accurate and convenient monitoring of human blood pressure. A camera-based non-contact human IPPG signal acquisition system is designed. The system can perform experimental acquisition under ambient light, effectively reducing the cost of non-contact pulse wave signal acquisition while simplifying the operation process.

4.4.3 Sleep Staging

Figure 1. Measurement setup. Three cameras positioned between two illumination units were placed above the bed to continuously capture the patient's face during the overnight recording.



- [Contactless Camera-Based Sleep Staging: The HealthBed Study]
- We evaluated a camera-based remote photoplethysmography (PPG) setup to perform automated classification of sleep stages in near darkness. Based on the contactless measurement of pulse rate variability, we use a previously developed HRV-based algorithm for 3 and 4-class sleep stage classification.

5. Oxygen controlling

- The Smart Bed is equipped with an oxygen control system that monitors and regulates oxygen levels pulsing with the use breath pattern.
- This feature ensures optimal oxygen supply during sleep, enhancing overall comfort and sleep quality.

5. Oxygen controlling



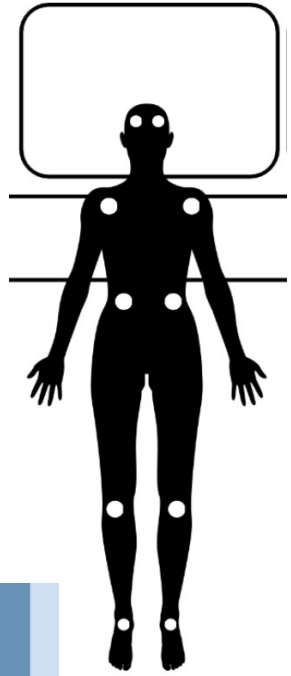
- Model: DEDAKJ DE-1A
- 7 liter per minute
- 96% purified oxygen

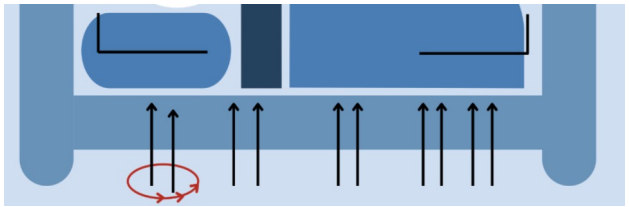
6. 10 movable lifting points for manipulating sleep position

- We understand that everyone's comfort needs are different, so the Smart Bed offers 10 movable lifting points.
- These points enable precise adjustments to find the perfect sleep position, minimizing pressure points and promoting spinal alignment.

6. 10 movable lifting points for manipulating sleep position







- 10 servo lifter engines
- Use data from camera to track and fit 10 point to the user skeleton map
- Slowly lift up/down to manipulate user sleeping position.
- 50cm lift
- Push: 500N
- Movement: 7-12mm/s
- Noise: <42dB

7. Temperature & thermal environment

- Say goodbye to restless nights due to temperature discomfort with the Smart Bed's thermal control system.
- The bed's advanced technology regulates temperature to create a cozy sleep environment that suits your preferences.

7. Temperature & thermal environment







- Monoblock Electrical Cabinet Air Conditioner
  - Cooling Capacity 300W
  - Noise 55dB
  - Ambient Temperature Limits -5~43°C
- Use input source from thermal sensors to control the ambient temperature

8. Sound & Music

- Immerse yourself in a world of relaxation with the Smart Bed's integrated sound and music system.
- Choose from soothing melodies, binaural beats, or connect your favorite devices to set the perfect ambiance for sleep.

8. Sound & Music



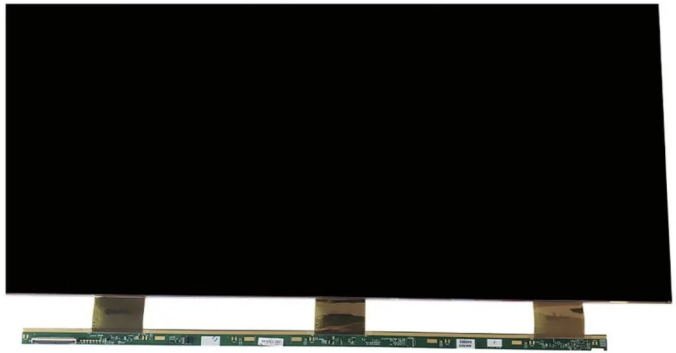
- 5.1 sound system: Denon SYS 2020

9. Light & Video

- Customizable ambient lighting helps create a calming atmosphere for bedtime and waking up.
- The Smart Bed also supports video integration, allowing for leisurely entertainment or visual sleep aids.

9. Light & Video





- Ambient RGB led system
- 45” 4k LCD panel to show content and controlling the chamber functions

10. Safety mechanism

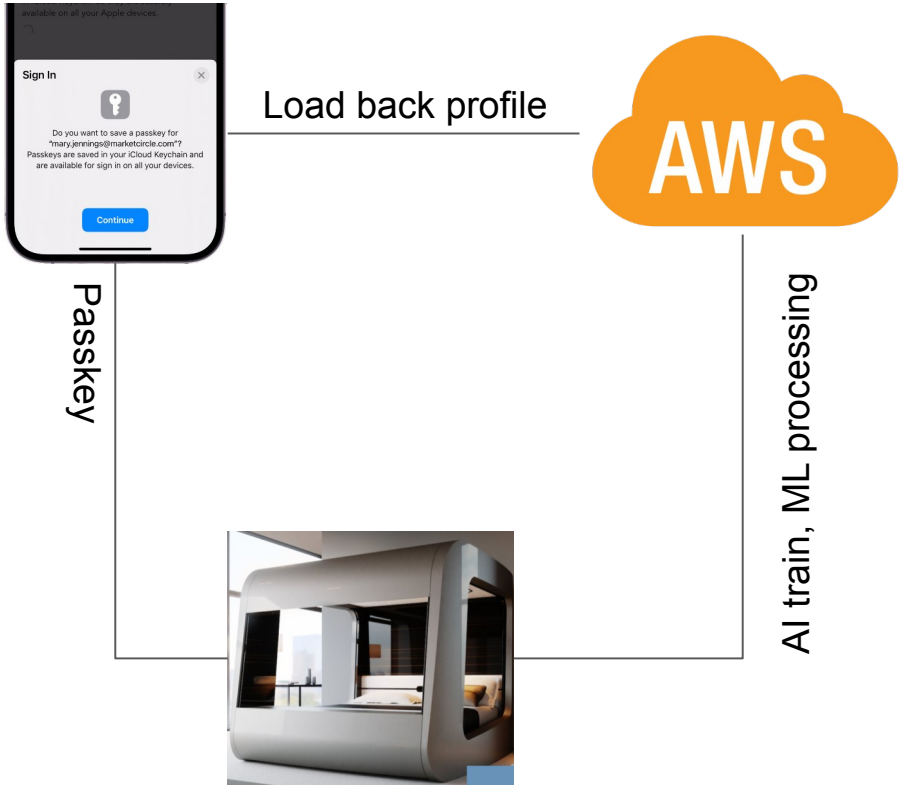






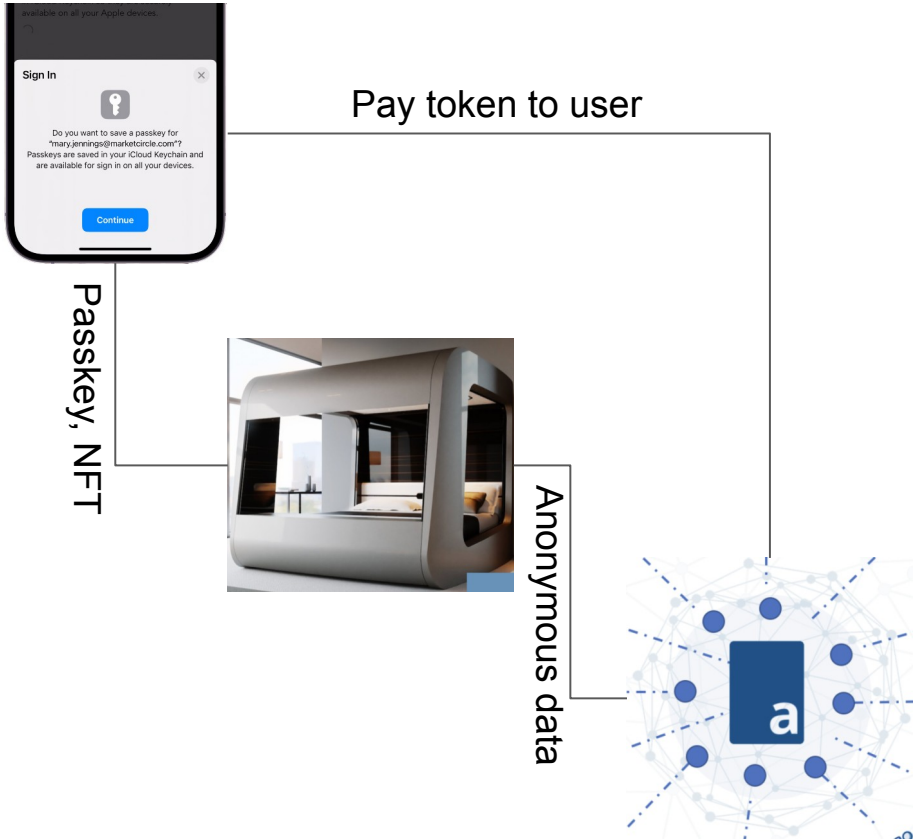
- Aqara Smoke Sensor JY-GZ-01AQ
- Mini Fire Stop 500ml.
- Smart Lock: Hafele ER5100
- The lock will auto open when emergency situation happen.

11. Cloud & Central AI



- User can opt in to stream anonymous data to the bed cloud service. The data is used to feed to continuously train AI brain.
- User can load back their profile to any smartbed to get the same setup suiting for their sleep.

12. User profile & data passkey & blockchain



- User registration by scanning a QR code using their smartphone.
- All the data will be hashed by their passkey and get approval by their smartphone biometric information
- We build and in house high performance, storage efficiency Amate private blockchain which is peering with some popular chain like ethereum, polkadot…
- User can opt in to provide their sleeping data to push to Amate data lake. The medical researchers who access to user's data must pay them some token.
- All the data in the Amate data lake is anonymous.

13. Release plan

Phase 1: MVP (V0)
(July 2023 - end of November 2023)

Product analysis
Business analysis
Setup backend server/transmission protocol, central controller board
Design 3D prototype (mechanical, kinematic, modeling)
Website, branding, art design
Manufacturing product frame
Adding living factor sensors
Testing additional parts - bed kinematic fundamental 10 points
Testing additional parts - air conditioner
Testing additional parts - oxygen input
Testing additional parts - video/screen content shower
Release init MVP version

13. Release plan

Phase 2: Launch V1
(December 2023 - end of March 2024)

Collect feedback/data from V0
Add improved living factor sensors
Collect data to local controller and push data to central server/dashboard
Upgrade and improve previous parts
Improve total product design
Release V1

13. Release plan

Phase 3: Launch V2
(April 2024 - mid of May 2024)

Collect feedback/data from V1
Add improved living factor sensors
Upgrade and improve previous parts
Data analysis
AI algorithms health check
AI algorithms personalized healthcare/content/bed position
Release V2

13. Release plan

Phase 4: Go To Market & Healthcare Partners
(mid of May 2024 - end of June 2024)

Improve AI for personalize setup
Website rebrand, add customer service/info functions
Build product care procedure/monitoring center/disaster resolving apartment
Build web portal/mobile app for client
Build web portal/mobile app for partners (doctors, healthcare center, pharmaceutical company)
Build dashboard for top central admin
Release Partner & Client portals

14. Literature review

A New Approach for Detecting Sleep Apnea Using a Contactless Bed Sensor: Comparison Study

Background: At present, there is an increased demand for accurate and personalized patient monitoring because of the various challenges facing health care systems. For instance, rising costs and lack of physicians are two serious problems affecting the patient's care. Nonintrusive monitoring of vital signs is a potential solution to close current gaps in patient monitoring. As an example, bed-embedded ballistocardiogram (BCG) sensors can help physicians identify cardiac arrhythmia and obstructive sleep apnea (OSA) non intrusively without interfering with the patient's everyday activities. Detecting OSA using BCG sensors is gaining popularity among researchers because of its simple installation and accessibility, that is, their non wearable nature. In the field of nonintrusive vital sign monitoring, a microbend fiber optic sensor (MFOS), among other sensors, has proven to be suitable. Nevertheless, few studies have examined apnea detection.
Objective: This study aims to assess the capabilities of an MFOS for nonintrusive vital signs and sleep apnea detection during an in-lab sleep study. Data were collected from patients with sleep apnea in the sleep laboratory at Khoo Teck Puat Hospital.
Methods: In total, 10 participants underwent full polysomnography (PSG), and the MFOS was placed under the patient's mattress for BCG data collection. The apneic event detection algorithm was evaluated against the manually scored events obtained from the PSG study on a minute-by-minute basis. Furthermore, normalized mean absolute error (NMAE), normalized root mean square error (NRMSE), and mean absolute percentage error (MAPE) were employed to evaluate the sensor capabilities for vital sign detection, comprising heart rate (HR) and respiratory rate (RR). Vital signs were evaluated based on a 30-second time window, with an overlap of 15 seconds. In this study, electrocardiogram and thoracic effort signals were used as references to estimate the performance of the proposed vital sign detection algorithms.
Results: For the 10 patients recruited for the study, the proposed system achieved reasonable results compared with PSG for sleep apnea detection, such as an accuracy of 49.96% (SD 6.39), a sensitivity of 57.07% (SD 12.63), and a specificity of 45.26% (SD 9.51). In addition, the system achieved close results for HR and RR estimation, such as an NMAE of 5.42% (SD 0.57), an NRMSE of 6.54% (SD 0.56), and an MAPE of 5.41% (SD 0.58) for HR, whereas an NMAE of 11.42% (SD 2.62), an NRMSE of 13.85% (SD 2.78), and an MAPE of 11.60% (SD 2.84) for RR.
Conclusions: Overall, the recommended system produced reasonably good results for apneic event detection, considering the fact that we are using a single-channel BCG sensor. Conversely, satisfactory results were obtained for vital sign detection when compared with the PSG outcomes. These results provide preliminary support for the potential use of the MFOS for sleep apnea detection.

14. Literature review

Multiparameter Sleep Monitoring Using a Depth Camera

In this study, a depth analysis technique was developed to monitor the user's breathing rate, sleep position, and body movement while sleeping without any physical contact. A cross-section method was proposed to detect the user's head and torso from the sequence of depth images. In the experiment, eight participants were asked to change the sleep positions (supine and side-lying) every fifteen breathing cycles on the bed. The results showed that the proposed method is promising to detect the head and torso with various sleeping postures and body shapes. In addition, a realistic overnight sleep monitoring experiment was conducted. The results demonstrated that this system is promising to monitor the sleep conditions in realistic sleep conditions and the measurement accuracy was better than the first experiment. This study is important for providing a non-contact technology to measure multiple sleep conditions and assist users in better understanding of his sleep quality.

14. Literature review (oxygen)

Contactless Camera-Based Sleep Staging: The HealthBed Study

Polysomnography (PSG) remains the gold standard for sleep monitoring but is obtrusive in nature. Advances in camera sensor technology and data analysis techniques enable contactless monitoring of heart rate variability (HRV). In turn, this may allow remote assessment of sleep stages, as different HRV metrics indirectly reflect the expression of sleep stages. We evaluated a camera-based remote photoplethysmography (PPG) setup to perform automated classification of sleep stages in near darkness. Based on the contactless measurement of pulse rate variability, we use a previously developed HRV-based algorithm for 3 and 4-class sleep stage classification. Performance was evaluated on data of 46 healthy participants obtained from simultaneous overnight recording of PSG and camera-based remote PPG. To validate the results and for benchmarking purposes, the same algorithm was used to classify sleep stages based on the corresponding ECG data. Compared to manually scored PSG, the remote PPG-based algorithm achieved moderate agreement on both 3 class (Wake–N1/N2/N3–REM) and 4 class (Wake–N1/N2–N3–REM) classification, with average κ of 0.58 and 0.49 and accuracy of 81% and 68%, respectively. This is in range with other performance metrics reported on sensing technologies for wearable sleep staging, showing the potential of video-based non-contact sleep staging.

14. Literature review (oxygen)

Effects of negative oxygen ions on sleep quality (PDF)

This paper aims to investigate the effects of negative oxygen ions on sleep quality. An experiment was held using both subjective and physiological methods on 33 subjects (16 females and 17 males). The subjective assessment on sleep quality was surveyed and the duration of every sleep stage was determined. The results of this experiment indicate that the negative oxygen ions have a significant effect on sleep quality. The subjective results indicate that with negative oxygen ions subjects fell asleep easier, meanwhile, the objective results indicate that the duration of sleep onset latency was shortened and the slow-wave sleep was lengthened. All these results show that negative oxygen ions can improve people's sleep quality.

Obstructive Sleep Apnea and Oxygen Therapy: A Systematic Review of the Literature and Meta-Analysis (PDF)

Hypoxemia is an immediate consequence of obstructive sleep apnea. Oxygen (O2) administration has been used as an alternative treatment in patients with obstructive sleep apnea (OSA) who do not adhere to continuous positive airway pressure (CPAP) in order to reduce the deleterious effects of intermittent hypoxemia during sleep. This systematic review aims to investigate the effects of O2 therapy on patients with OSA.

14. Literature review (Sleep position)

The Relationship between Sleeping Position and Sleep Quality: A Flexible Sensor-Based Study (PDF)

The use of flexible wearable sensors to monitor the impact of sleeping position and turning frequency on sleep and to study sleep patterns can help bedridden patients heal and recover. The flexible wearable sleeping-position monitoring device was designed and developed using a flexible angle sensor and a six-axis motion sensor to measure the dynamic changes in body posture during sleep. Based on the changes in the output parameters of the flexible angle sensor and the six-axis motion sensor, we determined the change in the subject's lying position, verifying and analyzing the relationship between lying position preference, turning frequency, and sleep quality in healthy subjects. The sleeping-position monitoring device was worn by 13 subjects (7 males and 6 females) without sleep disorders before the sleep experiment. They performed more than 50 sleeping-position changes to ensure the accuracy of the monitoring device. Subjects slept in their beds for 8 h per night for 15 nights. During that time, they wore the sleeping-position monitoring device and a wristband sleep-monitoring bracelet on their left hand, and gathered the subjective sleep data using questionnaires. The results show that the most critical influencing factors are sleeping-position preference and frequency of turning. Data analysis reveals that subjects with a preference for right-sided lying and a lower frequency of turning had better sleep quality.

14. Literature review (Thermal control)

Effects of thermal environment on sleep and circadian rhythm (PDF)

The thermal environment is a key determinant of sleep because thermoregulation is strongly linked to the mechanism regulating sleep. Excessively high or low ambient temperature (Ta) may affect sleep even in healthy humans without insomnia. Furthermore, disturbed nocturnal sleep affects not only daytime activities, but is also related to various adverse health effects, such as obesity, quality of life, and even mortality. These findings indicate that maintaining a comfortable thermal sleep environment is important for sleep maintenance as well as daytime activities and health status. Among the various thermal environmental factors, the relationship between Ta and the use of clothing and beddings differs greatly between humans and animals. The effects of Ta on sleep stages differ depending on whether subjects are semi-nude or use bedding and clothing, making it difficult to extrapolate the results of animal studies related to thermal environment and sleep to humans. The use of clothing and beddings greatly aids in maintaining the body temperature at an acceptable thermal state in a variety of environments by providing thermal resistance for the human body from its environment. In this review, based on our studies related to thermoregulation and sleep in humans, the effects of the thermal environment on sleep and circadian rhythm are discussed. The effects of heat and cold exposure, including bedding and clothing conditions along with their effects in the elderly, are of special interest, because these are the important factors that most impact sleep and thermoregulation.

14. Literature review (Sound & Music & Video)

Effects of pleasant sound on overnight sleep condition: A crossover randomized study (PDF)

It is desirable to improve sleep quality since poor sleep results in decreases in work productivity and increases in risks of lifestyle-related diseases. Sleep spindles in sleep EEG are waveforms that characterize non-REM sleep Stage 2 (Stage N2). Music therapy has been adopted as a non-pharmacological therapy for sleep quality improvement; however, few studies mention the relationship between music during sleep and spindles. We conducted a crossover randomized study to investigate music's effects on spindles and sleep parameters. Polysomnography (PSG) was performed on 12 adult males with sleep difficulties over three nights, during which they were exposed to three different acoustic environments–silent, white noise, and pleasant sounds–throughout the night, in a crossover randomized setting. Half of the participants with large WASO were defined as the sleep maintenance difficulty group. We investigated whether pleasant sounds shortened sleep onset latency (SOL) and increased the number of spindles (SN) and spindle density (SD) compared to white noise, using silence as the reference. The spindles were detected using the previously reported automatic spindle detection algorithm. After one patient was excluded due to data corruption, a total of 11 participants, including the sleep maintenance difficulty group (n = 5), were analyzed. For all participants, SOL was not significantly shorter with pleasant sound than with white noise ($p = 0.683$); for the sleep maintenance difficulty group, SOL tended to be shorter with pleasant sound than with white noise ($p = 0.060$). Compared to white noise, the SN increased in pleasant sound for 7 of 11 (4 of 5 in the sleep maintenance difficulty group), and SD increased for 5 of 11 (3 of 5 in the sleep maintenance difficulty group). The results suggest that all-night background sound exposure may affect SN and SD. Future research should investigate whether background sound exposure reduces sleep-related distress, achieves sound sleep, or improves daytime psychomotor function.

14. Literature review (Sound & Music & Video)

EFFECT OF BACKGROUND NOISE ON SLEEP QUALITY (PDF)

Indoor and outdoor noise are major causes of sleep disruption in western countries. Previous research has shown that administration of constant white noise may improve sleep quality by increasing the acoustic arousal threshold. Small trials and anecdotal findings suggest that white noise may also influence the brain electrical activity and improve sleep quality by reducing sleep onset latency and promoting a deeper sleep.

14. Literature review (Sound & Music & Video)

The music that helps people sleep and the reasons they believe it works: A mixed methods analysis of online survey reports (PDF)

Sleep loss is a widespread problem with serious physical and economic consequences. Music can impact upon physical, psychological and emotional states, which may explain anecdotal reports of its success as an everyday sleep aid. However, there is a lack of systematic data on how widely it is used, why people opt for music as a sleep aid, or what music works; hence the underlying drivers to music-sleep effects remain unclear. We investigated music as a sleep aid within the general public via a mixed methods data online survey (n = 651) that scored musicality, sleep habits, and open text responses on what music helps sleep and why. In total, 62% of respondents stated that they used music to help them sleep. They reported fourteen musical genres comprising 545 artists. Linear modeling found stress, age, and music use as significant predictors of sleep quality (PSQI) scores. Regression tree modeling revealed that younger people with higher musical engagement were significantly more likely to use music to aid sleep. Thematic analysis of the open text responses generated four themes that described why people believe music helps sleep: music offers unique properties that stimulate sleep (Provide), music is part of a normal sleep routine (Habit), music induces a physical or mental state conducive to sleep (State), and music blocks an internal or external stimulus that would otherwise disrupt sleep (Distract). This survey provides new evidence into the relationship between music and sleep in a population that ranged widely in age, musicality, sleep habits and stress levels. In particular, the results highlight the varied pathways of effect between music and sleep. Diversity was observed both in music choices, which reflected idiosyncratic preferences rather than any consistent musical structure, and in the reasons why music supports good sleep, which went far beyond simple physical/mental relaxation.

14. Literature review (Sound & Music & Video)

Music improves sleep quality in students (PDF)

This paper is a report of a study to investigate the effects of music on sleep quality in young participants with poor sleep. Sleep disorders may result in fatigue, tiredness, depression and problems in daytime functioning. Music can reduce sympathetic nervous system activity, decrease anxiety, blood pressure, heart and respiratory rate and may have positive effects on sleep via muscle relaxation and distraction from thoughts. Control groups have not been used in most previous studies. We used a three-group repeated measures design. Ninety-four students (aged between 19 and 28 years) with sleep complaints were studied in 2006. Participants listened for 45 minutes either to relaxing classical music (Group 1) or an audiobook (Group 2) at bedtime for 3 weeks. The control group (Group 3) received no intervention. Sleep quality was measured using the Pittsburgh Sleep Quality Index before the study and weekly during the intervention. Depressive symptoms in experimental group participants were measured using the Beck Depression Inventory. Repeated measures revealed a main effect of TIME ($P < 0.0001$) and an interaction between TIME and GROUPS ($P < 0.0001$). Post hoc tests with Bonferroni correction showed that music statistically significantly improved sleep quality ($P < 0.0001$). Sleep quality did not improve statistically significantly in the audiobook and the control group. Depressive symptoms decreased statistically significantly in the music group ($P < 0.0001$), but not in the group listening to audiobooks. Relaxing classical music is an effective intervention in reducing sleeping problems. Nurses could use this safe, cheap and easy to learn method to treat insomnia.

14. Literature review (Sound & Music & Video)

Binge Viewing, Sleep, and the Role of Pre-Sleep Arousal (PDF)

To investigate the prevalence of binge viewing, its association with sleep and examine arousal as an underlying mechanism of this association.

14. Literature review (Sound & Music & Video)

Effects of Music Videos on Sleep Quality in Middle-Aged and Older Adults With Chronic Insomnia: A Randomized Controlled Trial

Listening to soothing music has been used as a complementary therapy to improve sleep quality. However, there is no empirical evidence for the effects of music videos (MVs) on sleep quality in adults with insomnia as assessed by polysomnography (PSG). In this randomized crossover controlled trial, we compared the effects of a peaceful Buddhist MV intervention to a usual-care control condition before bedtime on subjective and objective sleep quality in middle-aged and older adults with chronic insomnia. The study was conducted in a hospital's sleep laboratory. We randomly assigned 38 subjects, aged 50-75 years, to an MV/usual-care sequence or a usual-care/MV sequence. After pretest data collection, testing was held on two consecutive nights, with subjects participating in one condition each night according to their assigned sequence. Each intervention lasted 30 min. Sleep was assessed using PSG and self-report questionnaires. After controlling for baseline data, sleep-onset latency was significantly shorter by approximately 2 min in the MV condition than in the usual-care condition ($p = .002$). The MV intervention had no significant effects relative to the usual care on any other sleep parameters assessed by PSG or self-reported sleep quality. These results suggest that an MV intervention may be effective in promoting sleep. However, the effectiveness of a Buddhist MV on sleep needs further study to develop a culturally specific insomnia intervention. Our findings also suggest that an MV intervention can serve as another option for health care providers to improve sleep onset in people with insomnia.

14. Literature review (Sound & Music & Video)

A Low-Noise, Non-Contact EEG/ECG Sensor (PDF)

Typical electroencephalogram (EEG) and electrocardiogram (ECG) sensors require conductive gel to ensure low-impedance electrical contact between the sensor and skin, making set-up time-consuming and long-term recording problematic. We present a gel-free, non-contact EEG/ECG sensor with an on-board electrode that is capacitively coupled to the skin. Active shielding of the high-impedance input significantly reduces noise pickup, and reduces variations in gain as a function of gap distance. The integrated sensor combines amplification, bandpass filtering, and analog-to-digital conversion within a 1 inch diameter enclosure. The measured input-referred noise, over 1-100 Hz frequency range, is 2μVrms at 0.2mm sensor distance, and 17μVrms at 3.2mm distance. Experiments coupling the sensor to the human scalp through hair and to chest through clothing produce clear EEG and ECG recorded signals.

14. Literature review (Sound & Music & Video)

Deep Learning-Based Non-Contact IPPG Signal Blood Pressure Measurement Research

In this paper, a multi-stage deep learning blood pressure prediction model based on imaging photoplethysmography (IPPG) signals is proposed to achieve accurate and convenient monitoring of human blood pressure. A camera-based non-contact human IPPG signal acquisition system is designed. The system can perform experimental acquisition under ambient light, effectively reducing the cost of non-contact pulse wave signal acquisition while simplifying the operation process. The first open-source dataset IPPG-BP for IPPG signal and blood pressure data is constructed by this system, and a multi-stage blood pressure estimation model combining a convolutional neural network and bidirectional gated recurrent neural network is designed. The results of the model conform to both BHS and AAMI international standards. Compared with other blood pressure estimation methods, the multi-stage model automatically extracts features through a deep learning network and combines different morphological features of diastolic and systolic waveforms, which reduces the workload while improving accuracy.